

Marks : 50

a. **Cardiac output**

KATHMANDU UNIVERSITY
October/November 2021

Subject: Physiology

Level : BDS

Paper : I

Exam : BDS I (new course)

Marks : 50

Time : 2 hour 30 minutes

Date : November 2, 2021

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

- ① structural channel
② act as carrier protein
③ act as enzyme
④ Transport substances, hormones
- Mem protein = Integral & peripheral protein

SECTION "B"
[25 marks]

1. List the functions of membrane proteins of a cell. Compare and contrast between simple and facilitated diffusion. Give an example of secondary active transport that occurs across cell membrane. 225
co-transport [2+2+2=6]
Counter → Na & Glucose Na & H⁺
2. Enumerate the steps involved in hemostasis? How do platelets help in clot retraction? Briefly explain the steps of extrinsic pathway of blood coagulation with the help of a flow chart. [2+1+2=5]
3. Describe the ionic basis of action potential. [5]
4. Write short notes on: [3×3=9]
 - a. Hemostasis.
 - b. Three important roles of Na⁺-K⁺ pump.
 - c. Cyclic AMP second messenger pathway.

SECTION "C"
[25 marks]

5. Describe four forces influencing fluid movement for extracellular fluid distribution across the capillary wall. [6]
6. Describe briefly the different lung volumes and capacities. [5]
7. Define cardiac output. Describe in brief about the factors affecting cardiac output. [1+4=5]
8. Write short notes on: [3×3=9]
 - a. Chemical regulation of respiration.
 - b. Local metabolic changes that influence arteriolar radius.
 - c. Types of hypoxia.

INSTRUCTIONS TO THE CANDIDATE

Answer each section in a separate answer book
Give brief, relevant and legible answers
Illustrate your answer with well labeled diagrams whenever required
All questions are compulsory

SECTION "B"
[25 marks]

Classify passive and active transport with one example of each. [3+3=6]

Describe the stages of erythropoiesis with the help of well labeled diagram. [5]

Explain the role of sodium and potassium ions in the generation of different phases of action potential with the help of well labeled diagram. List any two differences between graded potential and action potential. [3+2=5]

Write short notes on:

- Gap junction.
- Effect of sympathetic stimulation on the heart.
- Role of T- helper cells in immune activation.

[3×3=9]

SECTION "C"
[25 marks]

Describe the various short term, intermediate and long term mechanisms for regulation of blood pressure. [6]

Explain Haldane effect with the help of a diagram. Mention its significance. [4+1=5]

What is hypoxia? State its types and explain hypoxic hypoxia in detail. [1+1+3=5]

[3×3=9]

Write short note on:

- Conducting system of heart.
- Chemical regulation of respiration.
- Vital capacity.

KATHMANDU UNIVERSITY
February/March 2020

Level : BDS

Subject: Physiology

Exam : BDS I

Paper : I

Time : 2 hour 30 minutes

Marks : 50

Date : March 2, 2020

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"
[25 marks]

1. Describe erythropoiesis. Name the essential factors required and their role. [4+2=6]
2. Describe different types of membrane transports by giving one example of each. [5]
3. Name the cells found in the blood. List the white blood cells in the blood and give their functions. [1+4=5]
4. Write short notes on: [3×3=9]
 - a. Negative feedback with an example
 - b. Macrophage
 - c. Presynaptic ganglia in autonomic fibres

SECTION "C"
[25 marks]

5. What is the duration of cardiac cycle? Name the phases of cardiac cycle. Explain with help of labelled diagram, the phases in proper sequence. [1+2+3=6]
6. Define the vital capacity. Name the physiological factors affecting it and describe its relevance with timed vital capacity. [1+3+1=5]
7. Explain the baroreceptor reflex in relation to short-term regulation of blood pressure. [5]
8. Write short notes on: [3×3=9]
 - a. Acclimatization
 - b. Heart sounds
 - c. Dead space

KATHMANDU UNIVERSITY
July/August 2018

Subject: Physiology

Paper : I

Marks : 50

Level : BDS
Exam : BDS I (new course)
Time : 2 hour 30 minutes
Date : July 11, 2018

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"
[25 marks]

1. Define action potential. Illustrate five properties of action potential in detail. [1+5=6]
2. List any four functions of plasma proteins. Explain the function of albumin for maintenance of colloidal osmotic pressure across the capillary wall with appropriate diagram. [2+3=5]
3. Write the effects of sympathetic stimulation in different organs and explain how these effects are useful in 'fight or flight' responses. [3+2=5]
4. Write short note on:
 - a. Extrinsic pathway of blood clotting
 - b. Rh blood group
 - c. Functions of membrane proteins[3×3=9]

SECTION "C"
[25 marks]

5. Describe the changes in volume and pressure in the left ventricle during a cardiac cycle. (10/2) [3+3=6]
6. What is 'oxy-hemoglobin dissociation curve'? What information is obtained from the curve? List the factors affecting the curve and also their effect. [1+1+3=5]
7. Depict and describe a well labelled diagram for changes in intrapulmonary and intra-pleural pressures during inspiration and expiration. [3+2=5]
8. Write short note on:
 - a. Frank-Starling law of the heart muscle
 - b. Control of heart rate
 - c. Carbon dioxide transport in blood[3×3=9]

KATHMANDU UNIVERSITY
July/August 2019

Level : BDS

Subject: Physiology

Exam : BDS I (new course)

Paper : I

Time : 2 hour 30 minutes

Marks : 50

Date : August 6, 2019

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"
[25 marks]

1. Compare and contrast between extrinsic and intrinsic pathways of coagulation of blood. [6]
2. What is Cotransport? Give the examples and explain the mechanisms with help of a diagram. [1+4=5]
3. Describe the ionic basis of action potential in detail. [5]
4. Write short notes on: [3×3=9]
 - a. Rh blood group
 - b. Platelets
 - c. Neutrophil

SECTION "C"
[25 marks]

5. Explain the meaning, mechanism, and implication of the Frank-Starling law of the heart muscle. [2+3+1=6]
6. Define hypoxia. List types of hypoxia and its physiological basis. [1+2+2=5]
7. Define blood pressure. Describe short term regulation of blood pressure. [1+4=5]
8. Write short notes on: [3×3=9]
 - a. Hering-Breuer inflation reflex (Apneustic center)
 - b. Capacitance vessels
 - c. Surfactant

KATHIMANDU UNIVERSITY
December 2018/January 2019

Subject: Physiology

Paper : I

Marks : 50

Level : BDS
Sem : BDS I (new course)
Time : 2 hour 30 minutes
Date : January 7, 2019

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"
[25 marks]

- Describe the phases of action potential in skeletal muscle with their ionic basis. Explain the property of refractory period of action potential. [4+2=6] ✓
- Describe the differences between passive and active transport mechanisms. [5] ○
- Draw a flow diagram for 'in vitro' clotting mechanism triggered by the charged-wettable surface such as glass. [5]
- Write short note on: [3×3=9]
 - Second messenger
 - Hemopoiesis in adult life
 - Autonomic nervous system

SECTION "C"
[25 marks]

- Define arterial blood pressure. Describe two major determinants of arterial blood pressure in detail. [2+2+2=6] ✓
- Explain the inspiratory phase of respiration by describing the muscles involved and pressure changes in the lungs. [5]
- List two medium for transport of O₂ in blood. Explain three different causes for hypoxic hypoxia with appropriate examples. [2+3=5]
- Write short note on: [3×3=9]
 - Medullary chemoreceptors in respiratory function ✓
 - Baroreceptor mechanism of regulating blood pressure ✓
 - Role of endothelial cells in local autoregulatory mechanisms of blood flow ✓

KATHMANDU UNIVERSITY
December 2017/January 2018

Subject: Physiology

Level : BDS

Exam : BDS I (new course)

Time : 2 hour 30 minutes

Date : December 27, 2017

Paper : I

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"
[25 marks]

1. What is erythroblastosis fetalis? List the consequences of mismatched blood transfusion. [2+3=5]
2. Describe the ionic basis of action potential. (What is excitatory and inhibitory post-synaptic potential?) [3+3=6]
3. List the methods of transport of substances across the cell membrane. Describe the facilitated diffusion with examples. [2+3=5]
4. Write short note on: [3×3=9]
 - a. White blood cells
 - b. Gap-junction
 - c. List the sympathetic ganglia present in humans. How is heart rate effected by sympathetic nervous system?

SECTION "C"
[25 marks]

5. Define mean blood pressure and pulse pressure giving their normal values. Describe the factors which regulate the blood pressure in the normal range. [2+4=6]
6. What is hypoxia? Explain the various types of hypoxias, with examples. [2+3=5]
7. Describe the location and functions of peripheral chemoreceptors for regulation of respiration. [2+3=5]
8. Write short notes on: [3×3=9]
 - a. Starling law of heart
 - b. Surfactant
 - c. ST segment of ECG

KATHMANDU UNIVERSITY
May/June 2016

Subject: Physiology

Paper : I

Marks : 50

BDS

BDS I (new course)

2 hr. 30 minutes

June 5, 2016

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
Write brief, relevant and legible answers
Illustrate your answer with well labeled diagrams whenever required
All questions are compulsory

SECTION "B" [25 marks]

1. Define resting cell membrane potential. Describe ionic basis of action potential. [2+4=6]

2. Differentiate between passive and active transport. What is facilitated diffusion? [2+3=5]

3. Describe the stages of erythropoiesis. State the factors regulating erythropoiesis. [2+3=5]

4. Write briefly about:

- a. ABO grouping
- b. Action of parasympathetic nervous system on pupils, GIT and lung.
- c. Homeostatic regulation

[3×3=9]

SECTION "C" [25 marks]

5. Define cardiac cycle. Illustrate with diagram showing different events and time relations. [2+4=6]

6. Define the respiratory centers. Explain the mechanism of regulation of respiration. [2+3=5]

7. Describe the O₂ transport in the blood and the factors regulating O₂ dissociation curve. [2+3=5]

8. Write briefly about:

- a. SA node
- b. Coronary circulation
- c. Surfactant

[3×3=9]

KATHMANDU UNIVERSITY
May/June 2016

Level : BDS
Exam : BDS I
Time : 3 hrs.
Date : June 5, 2016

Subject: Physiology

Paper : I

Full marks: 70

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "A" [35 marks]

1. Describe the modern concept of transmission of nerve impulse through a synapse. Give a brief idea about the drugs acting on synaptic transmission. [5+2=7]
2. Mention the effects of sympathetic and parasympathetic system on: [5×1=5]
 - a. SA node
 - b. Coronary arteriols
 - c. Bronchial muscle
 - d. Ureters
 - e. Urinary bladder
3. Define vital capacity. State its normal value. How it is determined? How does it differ from timed vital capacity? What is the clinical significance of timed vital capacity? [1+1+1+1+3=7]
4. How oxygen is transported from lung to the tissue? What is Bohr effect? [4+2=6]
5. Write short notes on: [5×2=10]
 - a. High altitude sickness
 - b. Dysbarism
 - c. Generator potential
 - d. Hypoxic hypoxia
 - e. Electron microscopic structure of skeletal muscle

SECTION "B" [35 marks]

6. What are the proteins present in plasma? What are the origins of plasma proteins? State their functions. What is plasmapheresis? [1+1+3+2=7]
7. Describe the process of short term regulation of blood pressure after haemorrhage. What is hypovolemic shock? [5+2=7]
8. Write short notes on: [3×3=9]
 - a. Erythroblastosis foetalis
 - b. Functions of neutrophil
 - c. Role of baroreceptors in the regulation of vascular tone
9. What is cardiac cycle? Describe the different stages of cardiac cycle with diagram. [1+6=7]
10. Write briefly on: [2×2.5=5]
 - a. Peculiarities of cerebral circulation
 - b. Effect of "g" on blood pressure

BDS

BDS I (old course)

3 hrs.

September 14, 2015

Subject: Physiology

Paper : I

Full marks: 70

INSTRUCTIONS TO THE CANDIDATE

Answer each section in a separate answer book

Write brief, relevant and legible answers

Illustrate your answer with well labeled diagrams whenever required

All questions are compulsory

SECTION "A" [35 marks]

Name the types of cell junctions. Name the important location and function of each type of cell junction. [1.5+1.5+4=7]

Mention the site of origin of sympathetic nervous system. Name the important neurotransmitters released at pre-ganglionic and post ganglionic fibres of sympathetic nerves. Enumerate the structures innervated by sympathetic nerves in the heart and its function in each case. [1+1+3=5]

Briefly describe the location, structure and mechanism of stimulation of peripheral chemoreceptors. Add a note on their role in the regulation of respiration. [1+1+2+3=7]

List the factors which influence the rate of diffusion of oxygen in the lung. Describe the carriage of oxygen from the lungs to the tissues. [2+5=7]

Write briefly about:

- Resting membrane potential
- High altitude acclimatization
- Lung compliance

[3×3=9]

SECTION "B" [35 marks]

Define innate immunity. List the important mechanisms in the body which provide innate immunity. Briefly describe the role of blood cells in innate immunity. [1+2+4=7]

List the steps involved in extrinsic system of coagulation in the format of a flow diagram. [5]

Name the location, structure and innervation of arterial baroreceptors. Briefly describe their role in the regulation of arterial blood pressure. [3+4=7]

Briefly outline the regulation of blood flow to the skeletal muscles. How is increase in oxygen supply to the skeletal muscle brought about during isotonic exercise? [3+4=7]

[3×3=9]

Write briefly about:

- Coronary blood flow
- Role of calcium in the heart
- Heart sounds

KATHMANDU UNIVERSITY
2016

Subject: Physiology

Paper : 1

Marks : 70

BDS

BDS I

3 hour

October 21, 2016

INSTRUCTIONS TO THE CANDIDATE

Answer each section in a separate answer book
Write brief, relevant and legible answers
Illustrate your answer with well labeled diagrams whenever required
All questions are compulsory

SECTION "A" [35 marks]

Briefly outline the mechanism of development and maintenance of resting membrane potential in an excitable cell. [3+2=5]

State the sites of origin of parasympathetic nervous system. Name the neurotransmitters released at different levels and describe the effects of its stimulation on respiratory system. [1+1+3=5]

State the source, composition and functions of pulmonary surfactant. [1+1+3=5]

What is timed vital capacity? Explain its role in the diagnosis of pulmonary disease. [2+3=5]

Write briefly on: [3×5=15]

- High altitude acclimatization
- Peripheral chemoreceptors
- CO₂ transport

SECTION "B" [35 marks]

Explain the role of lymphocytes in immunity. [5]

What is erythrocyte sedimentation rate? How is it determined? Explain its role in the diagnosis and prognosis of disease. [1+1+3=5]

Briefly describe the origin and spread of cardiac impulse. State the effects of autonomic nervous system on them. [3+1+1=5]

Define cardiac output. Describe one method used to determine it. Name two conditions which cause an increase in cardiac output. Why? [1+2+1+1=5]

Write briefly on: [3×5=15]

- Circulatory shock
- Coronary circulation
- Filtration across capillary wall

KATHMANDU UNIVERSITY 2016

Subject: Physiology

Paper : 1

Marks : 50

BDS
BDS I (new course)
2 hour 30 minutes
October 21, 2016

INSTRUCTIONS TO THE CANDIDATE

Answer each section in a separate answer book
Give brief, relevant and legible answers
Illustrate your answer with well labeled diagrams whenever required
All questions are compulsory

SECTION "B" [25 marks]

Define homeostasis. List two different types of homeostatic regulatory mechanisms with appropriate examples. [2+2+2= 6]

What is resting membrane potential? Describe the ionic basis of action potential. [1+4=5]

What is erythropoietin? Illustrate the stages for red cell formation in bone marrow. [1+4= 5]

Write short note on:

- Functions of platelets
- Cell organelles
- Neurotransmitters released from preganglionic and postganglionic fibers in sympathetic nervous system [3×3= 9]

SECTION "C" [25 marks]

5. State Frank-Starling's law of heart. List the factors that increase the preload in heart. [2+4=6]

6. Define vital capacity. List pulmonary volumes, capacities and their normal values. [1+4= 5]

7. Briefly discuss the role of pontine and medullary centres in the neural control of respiration. [5]

[3×3=9]

8. Write short notes on:

- Windkessel effect
- Long term regulation of arterial BP
- Hypoxia